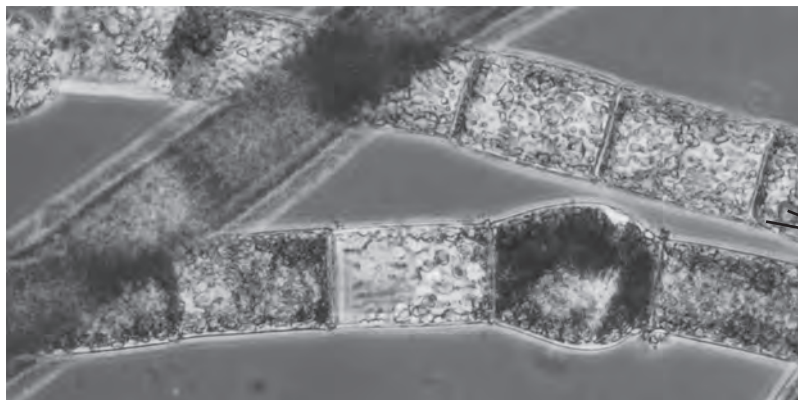


3 The three-domain system is used to classify organisms.

Photograph A shows some cells in one type of algae, as seen using one type of microscope.

The cells of this organism contain chloroplasts.



(Source: © Steve Taylor ARPS/Alamy Stock Photo)

Photograph A

(a) Name the domain which would contain this organism.

(1)

(b) Photograph B shows a chloroplast as seen using a different type of microscope.



(Source: © Science History Images/Alamy Stock Photo)

Photograph B

(i) Calculate the magnification of the chloroplast shown in photograph B.

Give your answer to **two** significant figures.

(2)

Answer



- (ii) Explain why the microscope used for photograph B shows more detail than the microscope used for photograph A.

(2)

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- (iii) Chloroplasts contain starch grains.

Which of the following is another structure that contains starch?

(1)

- ☐ **A** amyloplast
- ☐ **B** plasmodesmata
- ☐ **C** tonoplast
- ☐ **D** vacuole

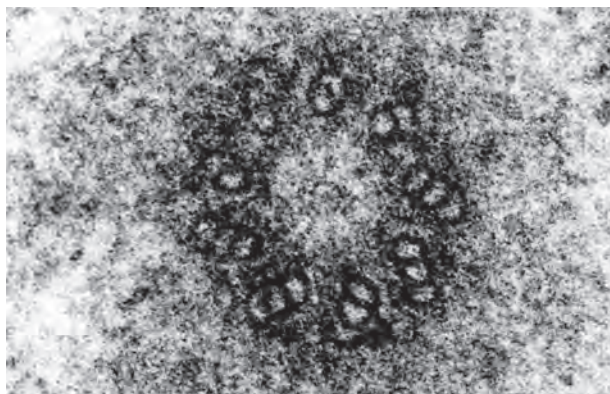
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- (c) The cells from algae contain some structures that are characteristic of animal cells.
The photograph shows one of these structures.



(Source: © DON W. FAWCETT/SCIENCE PHOTO LIBRARY)

The structure has the same function in the algal cell as in an animal cell.

Give the name and function of this structure.

(2)

Name

Function

(Total for Question 3 = 8 marks)

